

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Innovative AI/ML Engineer with over 9 years of experience in developing machine learning solutions, including large language model fine-tuning, clinical data processing, NLP pipelines, and production-grade model deployments. Deep expertise in transformer models, agentic frameworks like LangChain and LangGraph, MLOps with Docker/Kubernetes, and HIPAA-aware AI systems. Adept at translating complex clinical and operational needs into scalable and secure AI solutions in healthcare, IoT, and enterprise domains.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Frontend:** React, Redux, Next.js, Quasar (familiar), Tailwind CSS
- **Backend:** FastAPI, Flask, Node.js, Express.js
- **Data Processing & ML:** Pandas, NumPy, PyTorch, Scikit-learn, OpenCV
- **Database:** PostgreSQL, MySQL, SQLite, SQLAlchemy (ORM), MongoDB
- **DevOps/Tools:** Git, Docker, GitHub Actions, CI/CD, Vite, Webpack
- **Testing:** Pytest, Jest, React Testing Library
- **Other:** RESTful APIs, WebSockets, Agile/Scrum

WORK EXPERIENCE

Freelance AI/ML Consultant

Remote | Jan 2024 – Present

- Developed ML-powered features using PyTorch to analyze unstructured customer input data and trigger automated business rules, improving processing efficiency by 35%.
- Integrated a PyTorch-based classifier for detecting anomalies in transactional logs, enabling real-time alerts and reducing incident response times by over 50%.
- Trained custom neural networks using PyTorch Lightning and optimized training performance by batching, GPU acceleration, and model checkpointing.
- Created APIs to serve PyTorch inference models through FastAPI, ensuring secure, low-latency predictions within existing business workflows.
- Designed and documented an internal pipeline for retraining PyTorch models based on newly labeled user data, allowing continuous learning.
- Trained internal teams on building reproducible model pipelines using MLflow and custom experiment tracking solutions.

- Built automated E2E evaluation suites to track BLEU, ROUGE, and semantic match metrics for iterative model improvement.

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Designed and fine-tuned transformer-based LLMs (GPT, BERT) for healthcare tasks like discharge summary generation, medication extraction, and ICD-10 classification using hospital-specific datasets.
- Created LangChain-powered agentic workflows that automated EHR data ingestion, patient summarization, and diagnostic suggestion pipelines under HIPAA constraints.
- Developed secure data preprocessing pipelines that de-identified patient health records, encoded structured lab values, and aligned clinical notes for model training.
- Deployed containerized AI microservices using Docker and Kubernetes on private cloud infrastructure supporting real-time inference of NLP models with optimized latency.
- Led efforts in prompt engineering for few-shot learning configurations using OpenAI APIs to align outputs with clinical documentation standards.
- Built monitoring tools to evaluate ROUGE, BLEU, and medical accuracy metrics across different versions of fine-tuned LLMs for model performance audits.
- Integrated FastAPI-based model inference endpoints with hospital backend systems using FHIR-compliant APIs to securely handle PHI exchanges.
- Mentored junior developers and coordinated cross-functional teams in the implementation of multi-agent AI solutions for clinical workflow automation.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Developed end-to-end machine learning pipelines using PyTorch and TensorFlow to analyze and predict campaign engagement metrics and user behavior in healthcare marketing.
- Created NLP-driven chatbots for clinical pre-screening using Hugging Face transformers, capable of multilingual support and medical symptom triage.
- Orchestrated container-based deployments of AI models for inference using Docker and managed lifecycle workflows with GitHub Actions and MLflow.
- Applied knowledge distillation techniques to compress clinical intent detection models for low-latency mobile deployment.
- Integrated AI agents with CRM systems to automate follow-up actions based on user responses using RESTful APIs and asynchronous processing queues.
- Evaluated models using cross-validation and domain-specific KPIs, iteratively improving false-positive rates in intent classification and entity extraction tasks.
- Built scalable ETL systems to ingest healthcare marketing datasets, cleanse unstructured inputs, and support AI model training workflows.
- Documented complete model training procedures, including preprocessing scripts, inference logic, and deployment guidelines for team knowledge sharing.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Led development of AI tools for structuring unstructured EHR documents using named entity recognition and syntactic parsing pipelines.
- Built microservices for medical terminology standardization using UMLS datasets and clinical embedding generation with BioWordVec.
- Participated in creating a scalable LLM evaluation dashboard that visualized clinical summarization fidelity and supported error annotation feedback loops.
- Utilized Kubernetes for orchestrating multi-container deployments of training and inference jobs for LLMs across GPU nodes.
- Implemented automatic model retraining triggers based on EHR data drift analysis and inference monitoring logs.
- Collaborated with compliance teams to align ML system architecture with data privacy and PHI handling policies under HIPAA.
- Wrote modular Python code and documentation for clinical NLP components, ensuring reproducibility and auditability of model outputs.
- Contributed to weekly code reviews and architecture planning to scale clinical pipeline operations across hospital networks.

Software Developer

Decima Digital

November 2015 – April 2016

Tallinn, Estonia

- Built prototype NLP models to extract symptoms and medical history terms from free-text clinical interviews using spaCy pipelines.
- Integrated Kafka-enabled pipelines for streaming clinical event logs into real-time dashboard systems for EHR-based monitoring.
- Developed secure APIs to connect NLP models with external EMR platforms, ensuring anonymized, encrypted request handling.
- Created user interfaces for medical staff to provide feedback on AI-generated outputs, enabling active learning for summarization models.
- Tested knowledge distillation on custom BERT-based medical classification models, reducing size by 40% while retaining 90% accuracy.
- Implemented fuzzy matching algorithms to identify duplicated patient records based on inconsistent structured inputs.
- Participated in agile sprints to align data engineering teams with model-building requirements and data format transformations.
- Contributed Python libraries to internal clinical NLP toolkit used across multiple research projects.

Software Developer Intern

Borna Technology

September 2014 – August 2015

Tallinn, Estonia (On-Site)

- Assisted in annotation of clinical datasets for named entity recognition, supporting model training in symptom extraction pipelines.
- Developed Python scripts to automate keyword-based de-identification of free-text health notes as part of a HIPAA pilot project.
- Implemented early-stage Flask APIs for internal deployment of AI models on testbed infrastructure.

- Ran batch model evaluations using scikit-learn and compiled evaluation reports highlighting NLP misclassifications and drift.
 - Participated in weekly standups and learned practical deployment of ML prototypes into Dockerized services.
 - Contributed research on semantic search using TF-IDF and cosine similarity to help match patient notes to clinical guidelines.
 - Conducted exploratory analysis of hospital admission trends using Pandas, visualizing patterns with Matplotlib and Seaborn.
 - Supported integration tests of AI modules with mock EHR systems using synthetic health record data.
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EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ Microsoft Certified: Azure AI Engineer Associate
 - ❖ Hugging Face NLP Course Completion
 - ❖ Google Cloud Professional ML Engineer (In Progress)
 - ❖ Applied Data Science with Python Specialization – University of Michigan (Coursera)
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PROJECTS

✓ **AI-Generated Discharge Summaries for Hospital System**

Fine-tuned GPT-based models on de-identified hospital discharge notes

Built LangChain agent pipelines for patient history synthesis and quality checking

deployed on HIPAA-compliant cloud using Docker and monitored with custom metrics

✓ **Clinical Entity Extraction for EMR Analysis**

Built ClinicalBERT model pipeline for extracting medical conditions and medications

Integrated with EHR search interface for real-time information retrieval

✓ **PHI De-Identification Pipeline**

Developed regex/NLP hybrid approach to anonymize clinical free-text

Validated against hospital gold-standard datasets with over 98% recall